

List of Changes — Manuscript XQ11175

“How Focused Are LLMs? A Quantitative Study via Repetitive Deterministic Prediction Tasks”

All changes respond to specific referee comments (R1 = first referee, R2 = second referee). No content was added beyond what the referees requested.

Abstract

- Softened the description of the statistical model to an “effective” description linking correlated token errors to sequence-level failure (removed the implied first-principles attention mechanism). [R2-1]

Introduction

- Rewrote the “factorizable / product of per-step accuracies” statement to be precise: the factorization is exact only for site-local tasks (letter replacement, Pauli multiplication); integer addition has long-range carry dependence and is only approximately factorizable. [R1-1]
- Added a sentence framing the spin-glass analogy as an effective, phenomenological description, not a microscopic theory of the network. [R1-3, R2-1]

Experiment (Methods)

- Stated the sample size: up to $n = 100$ independent instances per length (10 repetitions $\times$ batch 10); noted that a subset of grok-4 runs use fewer repetitions ($n \sim 20\text{--}30$). [R1-2, R2-3]
- Stated that all accuracy curves report SAR with 95% confidence-interval error bars: bootstrap over the repeated batches for the geometric-mean SAR, and Wilson score intervals where SAR is the instance-level success rate (integer addition and the scaling collapse). [R1-2, R2-3]
- Strengthened the no-tools statement: all queries used plain text completion/chat calls with no tools, function-calling, or code-interpreter enabled; results reflect in-context computation only. [R2-4]
- Stated the decoding settings (temperature 0 where exposed; reasoning models use fixed internal sampling; per-model settings in the released code). [R1/R2 general]

Figures

- All accuracy-vs-length figures (cyclic letter replacement, integer addition, Pauli multiplication) now display 95% confidence intervals (bootstrap for the geometric-mean SAR curves; Wilson for the instance-level success rate, i.e. integer addition and the scaling collapse) and state n in the captions. [R1-2, R2-3]
- Added a new scaling-collapse figure: SAR vs N/N_* collapses onto a single crossover curve across all models and tasks, confirming the universal finite-size crossover. [R2-2]
- Phase (correlation-error) maps regenerated from the updated fits.

Analysis (Empirical formula)

- Added a model-comparison statement: an AIC comparison against the single-exponential ($\alpha = 1$) baseline favors the double-exponential in all but the highest-accuracy cases, which sit in the

$\alpha \rightarrow 1$ limit; the per-curve comparison is in the released code. [R2-5]

Analysis (Hypothesis / statistical model)

- Reframed the couplings J_{ij} as an effective form inspired by (not derived from) attention; removed causal “attention-induced/attention-mediated” language. [R2-1]
- Added an argument that disordered couplings are necessary: a field-only model (uniform or random h_i , no coupling) factorizes $\rightarrow$ single exponential, no crossover; uniform ferromagnetic couplings give a smooth transition, not the observed sharp, sample-to-sample crossover. [R1-4]
- Added clarification on symmetry and energetics: the field h breaks the $s \rightarrow -s$ symmetry (no $\mathbb{Z}_2$ degeneracy); at large N the disorder energy ($\sim N^2$) overtakes the field, so an incorrect configuration can become energetically competitive; the frustration is the competition between random-sign couplings and the ordering field. [R1-3]
- Added a sentence introducing the scaling-collapse prediction and referencing the new figure. [R2-2]
- Listed microscopic connection (extract J_{ij} from attention; perturb attention sparsity) as future work. [R2-1]

Divide-and-conquer section

- Stated that the divide-and-conquer prediction follows from the empirical accumulation law and does not depend on the SK-specific equations. [R1-5a]
- Attributed the Pro/Flash difference to per-call overhead and no longer present the Flash curve as a success. [R1-5b]
- Flagged extension of the divide-and-conquer test to the other tasks as future work. [R1-5c]

Appendix A (SK derivation)

- Corrected the $\mathcal{O}(J_0^4)$ coefficient in the large-field reduction from $+2J_0^4(N-1)^2$ to $-4J_0^4(N-1)^2$. [R1-6, R2-6]
- Reconciled the two functional forms: the double-exponential is the exponentiation of the leading-order $\mathcal{O}(J_0^2)$ correction (a phenomenological closed form valid for small J_0 , large h), NOT a strict term-by-term resummation; the leading-order $(\alpha, \beta_0) \leftrightarrow (J_0, h)$ mapping is retained. [R1-6, R2-6]
- Fixed a broken equation reference (“Eq. (0)”) to the explicit token-accuracy relation.

Conclusion / Future work

- Added two future directions tied to referee suggestions: (i) extending divide-and-conquer to the other tasks; (ii) connecting the model to network internals (extracting J_{ij} from attention; testing attention-sparsity perturbations). [R1-5c, R2-1]

Notes

- The manuscript and response use gender-neutral terms throughout.

- No new experiments were run; all strengthening derives from re-analysis of the already-collected data, code-level clarification, and revised framing.